LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Temperature compensation of instrument mechanisms: - Bimetal strip method, thermo resistance method, thermo magnetic shunt method.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Temperature compensation of instrument mechanisms: - Bimetal strip method, thermo resistance method, thermo magnetic shunt method.
5. Specific Objectives	15.1.1.3 Trainee will be able to explain the method used for temperature compensation in an instrument.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of instrument diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary cause of instrument error due to ambient temperature changes?
(b) Why is temperature compensation necessary for accurate flight instrument readings?
(c) List the three methods of temperature compensation covered in this lesson.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Bimetal strip method principle • Thermo resistance method operation • Thermo magnetic shunt method function	L & Ds	PPT, IB	15 min
(b)	• Comparison of compensation mechanisms • Application in specific instruments • Advantages and limitations of each method	L & Ds	PPT, PDF	20 min
(c)	• Summary of temperature compensation concepts • Key terminology review • Preparation for next lesson demonstration	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the bimetal strip method compensates for temperature variations.
(b) Describe the role of the thermo magnetic shunt in instrument accuracy.

12. Home Assignment and Follow Up	Review LP 02 syllabus and identify how temperature compensation concepts relate to the basic instrument mechanisms discussed previously.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Demonstration of mechanism of typical instrument.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min